

Introduction to Demographic Methods

(7-8: Future; Digital e Computational Demography; Conclusions)

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Introduction to Demographic Methods



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- 1- Introduction. Why study Demography? ... Git, version control, course materials GitHub
- 2- Mortality
- 3- Fertility
- 4- Population projections
- 5- Migration
- 6- Kinship
- 7- The future, and Digital and Computational Demography
- 8- Wrap up + Description of examination
- 9- Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Future of demographic research and Digital and Computational
Demography (which already has 10+ years of history)

Future: Digital and Computational Demography

1. Thick vs. big data



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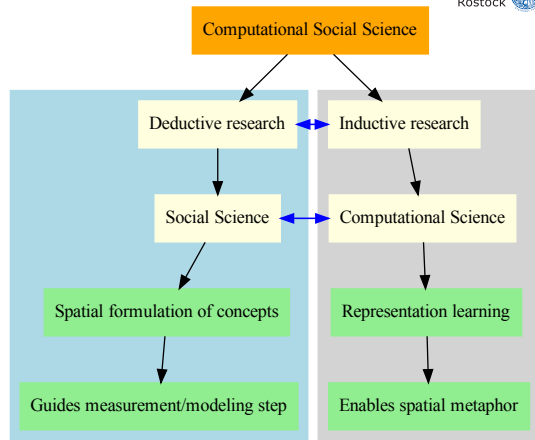
Future: Digital and Computational Demography

1. Thick vs. big data
2. Designed versus found data



Future: Digital and Computational Demography

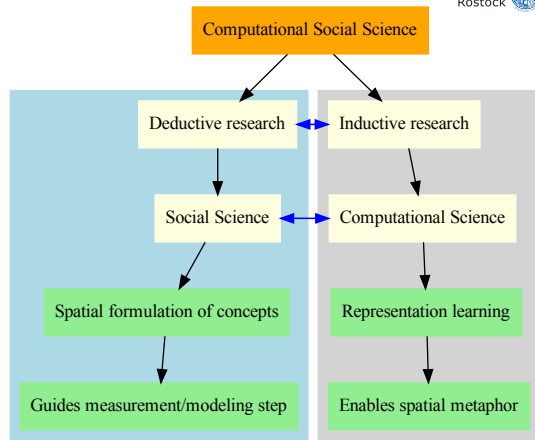
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2. Designed versus found data
3. Inductive vs. deductive approaches to research (see diagram on the right, and Akbaritabar (2024))



See more: Akbaritabar, 2024

Future: Digital and Computational Demography

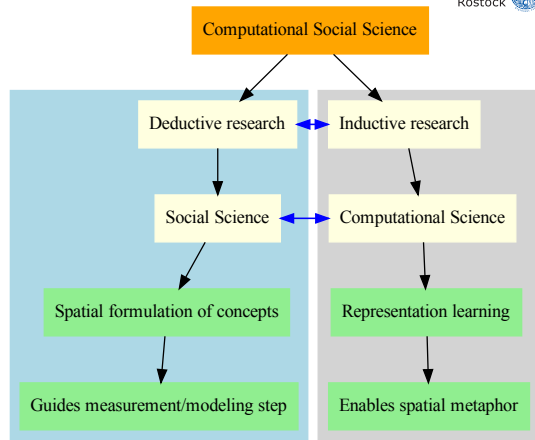
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2. Designed versus found data
3. Inductive vs. deductive approaches to research (see diagram on the right, and Akbaritabar (2024))
4. Cross-validation of approaches/methods; Hegemony of methodological frameworks (Castro Torres & Akbaritabar, 2024; Torres & Akbaritabar, 2025)



See more: Akbaritabar, 2024

Future: Digital and Computational Demography

1. Thick vs. big data
2. Designed versus found data
3. Inductive vs. deductive approaches to research (see diagram on the right, and Akbaritabar (2024))
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5. Mixed Methods



See more: Akbaritabar, 2024

Final exam

Please send me your slides for the 3 minutes presentation, at the latest, by 20:00 on the day before the session!

Where to learn basics of R and Python?

To start with Python

Check this website first:

https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:

<https://github.com/vtraag/intro-python>

Or this one by Data Carpentry:

<https://datacarpentry.org/python-ecology-lesson/>

To start with R

Check this website first:

<https://www.w3schools.com/r/default.asp>

This “very short introduction to R” is a good start:

<https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf>

Or this course by Data Carpentry:

<https://datacarpentry.org/R-genomics/index.html>

Reading materials for today



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Key reading material

-  Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2023, March). Digital and computational demography. In Research Handbook on Digital Sociology (pp. 48–86). Edward Elgar Publishing.

Additional reading materials

-  Salganik, M. J. (2018). Bit by bit: Social research in the digital age. Princeton University Press.
-  Kosinski, M., Matz, S. C., Gosling, S. D., Popov, V., & Stillwell, D. (2015). Facebook as a research tool for the social sciences: Opportunities, challenges, ethical considerations, and practical guidelines.. American Psychologist, 70(6), 543–556. <https://doi.org/10.1037/a0039210>

Thanks for your attention!

Questions and comments are always welcome!

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subject line: “[Demogr-SoSe-2026] your subject”

LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References

-  Akbaritabar, A. (2024). Thinking spatially in computational social science. *EPJ Data Science*, *13*(1), 1–12.
<https://doi.org/10.1140/epjds/s13688-023-00443-0>
-  Castro Torres, A. F., & Akbaritabar, A. (2024). The use of linear models in quantitative research. *Quantitative Science Studies*, 1–32.
https://doi.org/10.1162/qss_a_00294
-  Torres, A. F. C., & Akbaritabar, A. (2025). The hegemonic use of linear models in quantitative social sciences.
Bulletin of Sociological Methodology/Bulletin de Méthodologie Sociologique, 1–18. <https://doi.org/10.1177/07591063251349381>